

THI CUỐI KỲ 2 (2020-2021) PTTK HƯỚNG ĐỐI TƯỢNG

Học phần: Phân tích thiết kế hướng đối tượng - Học kỳ 2 năm học 2020-2021

Mã số sinh viên: 102160275

Họ tên: Nguyễn Lại Tam Vũ

Lớp: 16TCLC3

Nhóm: 19Nh13A

Thời gian bắt đầu: 06:56 25/08/2021

Thời gian nộp bài: 07:41 25/08/2021

Thời gian làm bài: 45:00

Tổng số câu hỏi: 40

Số câu làm đúng: 20

Số lần vi phạm: 0

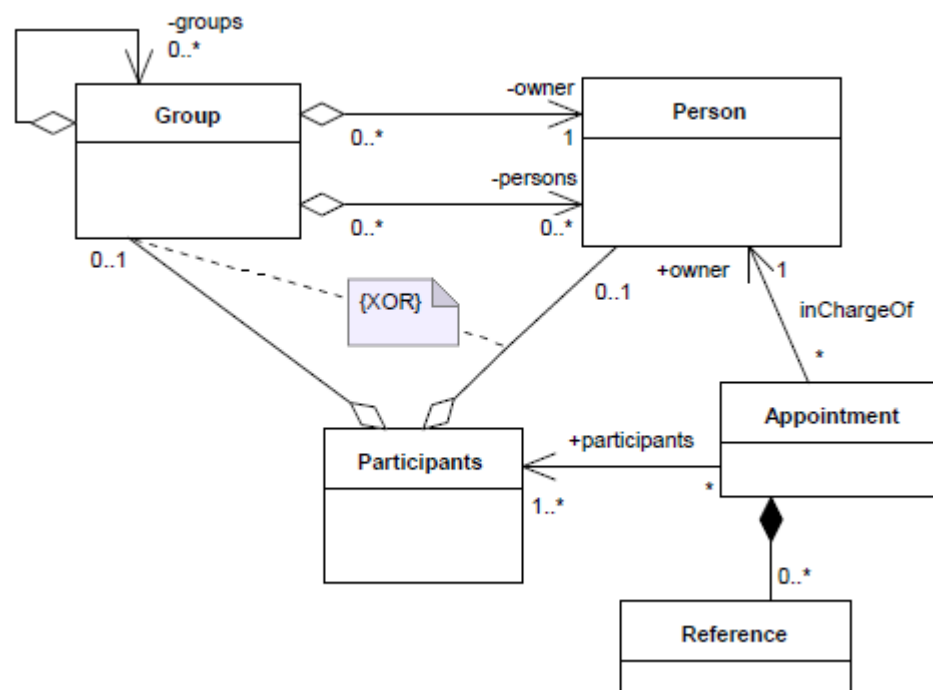
Số lần không nhận dạng được: 0

Trạng thái nhận dạng: Không nhận dạng

Điểm: 5

Câu 1 (Thời gian trả lời: 00:55)

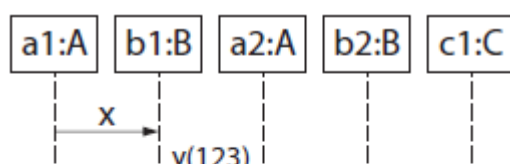
You are given the following clipping of a UML class diagram. Which of the following statements are true?

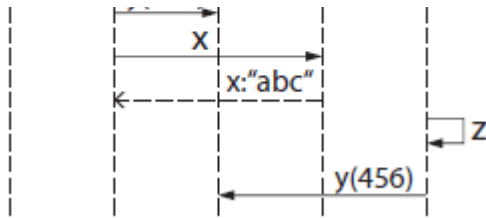


- ☐ A. Groups can only consist of persons, they cannot comprise of other groups.
- ☐ B. It is possible to navigate from a person to the appointments that he is in charge of.
- ☐ C. A participant that has been assigned to an appointment can be group and person at the same time.
- ☒ D. One group can consist of several persons.

Câu 2 (Thời gian trả lời: 00:55)

You are given the following sequence diagram. Which operations does class B have according to the diagram?





- ☐ A. z():void
- ☒ B. x():void
- ☒ C. x(void)
- ☐ D. y():int

Câu 3 (Thời gian trả lời: 01:00)

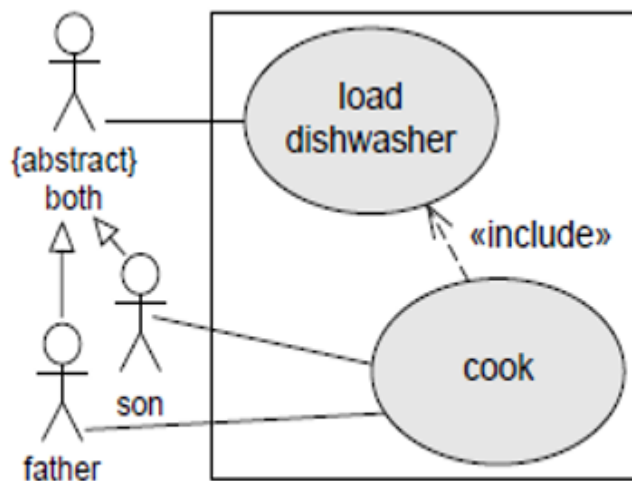
Which of the following statements are true ?

- ✓ ✓ A. If an object can only have one out of several associations, this may be modeled with a {xor} relationship
- B. You can specify the names of the roles the classes play in the relationship on the association ends
- ✗ ✓ C. A class must never have a relationship with itself
- D. A role name is noted in the upper left corner of a class symbol

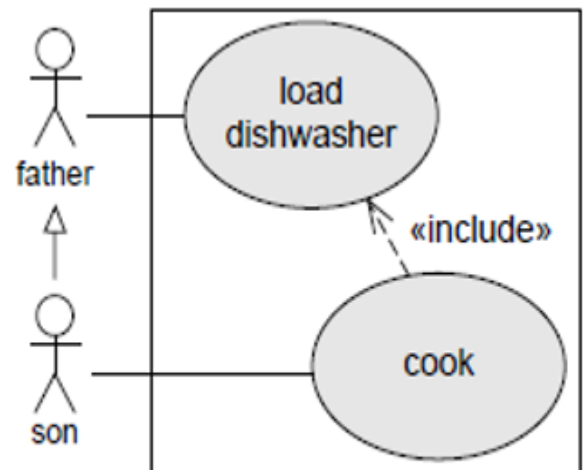
Câu 4 (Thời gian trả lời: 01:05)

How do you model the following situation with a UML use case diagram:

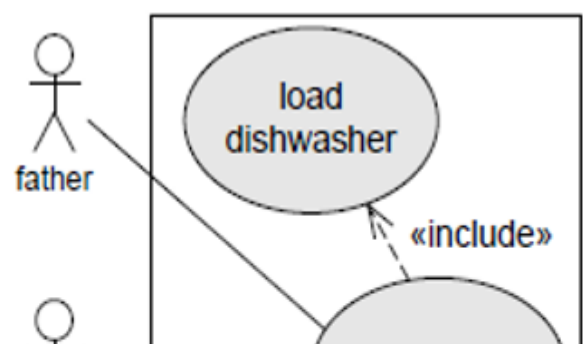
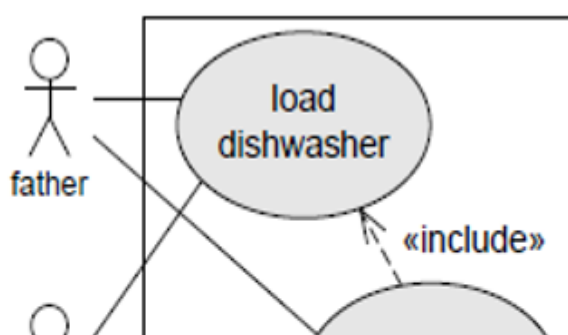
Father and son cook dinner. In the course of that, one of them always has to load the dishwasher. In this approach, only the son does the cooking.

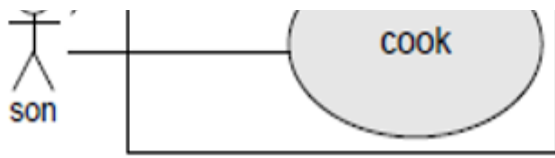


A

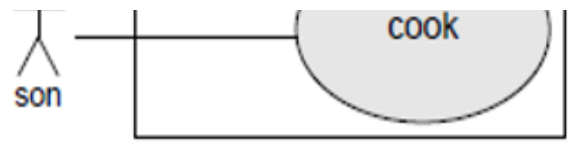


B





C

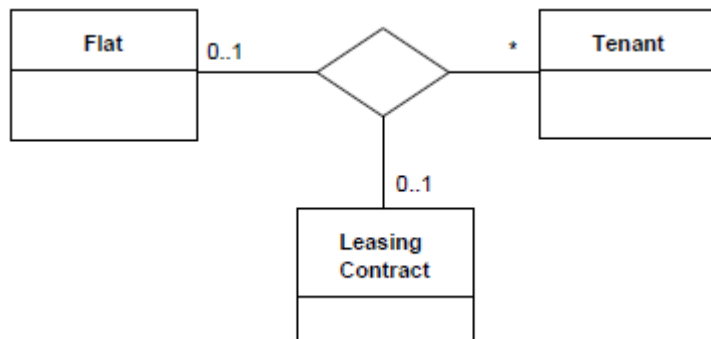


D

- ☒ A. A
☒ B. C
☐ C. D
☐ D. B

Câu 5 (Thời gian trả lời: 01:05)

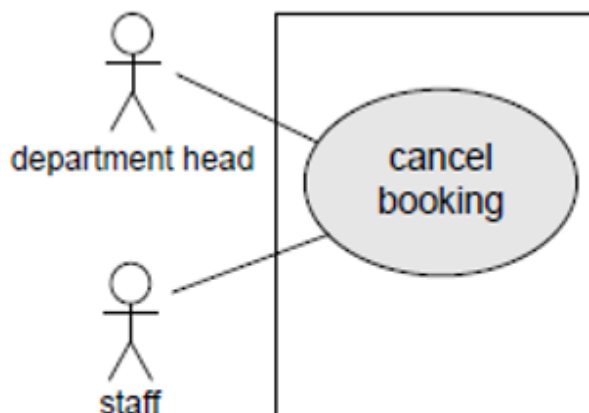
You are given the following clipping of a UML class diagram. Which of the following statements are true?



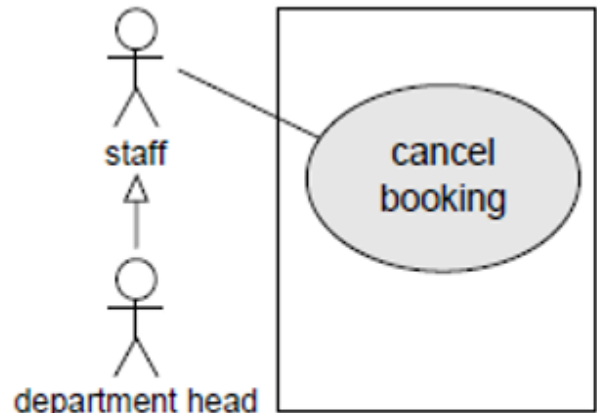
- A. One tenant can rent two flats with one leasing contract.
 ✓ ✓ B. One flat can be rented by multiple tenants with the same leasing contract.
 ✓ ✓ C. One tenant can rent multiple flats with different leasing contracts.
 D. One flat can be rented by multiple tenants with different leasing contracts.

Câu 6 (Thời gian trả lời: 01:15)

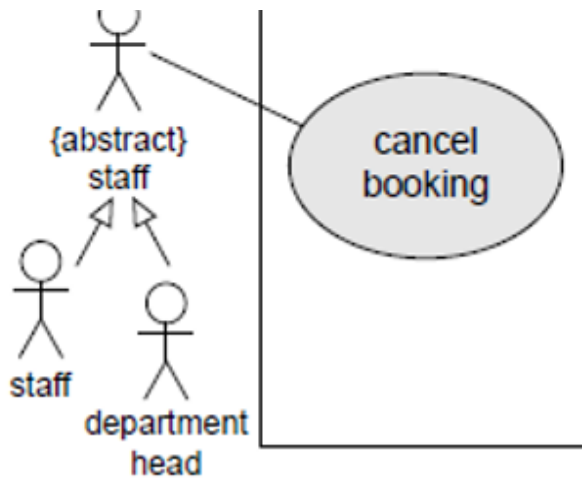
How do you model the following situation with a UML use case diagram: A travel booking is cancelled by a staff member or by the department head (who also is a staff member).



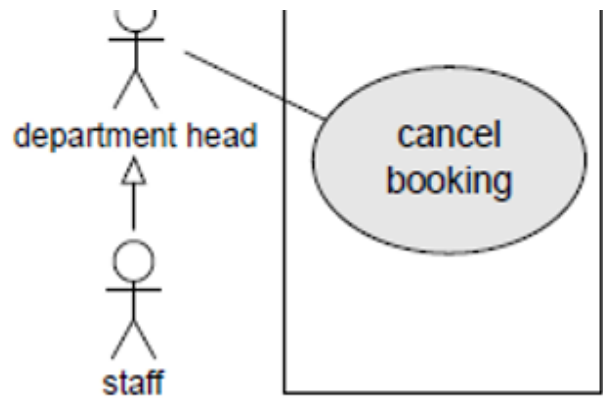
A



B



C

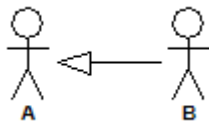


D

- ☐ A. C
- ☐ B. D
- ✓ ☒ C. B
- ☐ D. A

Câu 7 (Thời gian trả lời: 00:55)

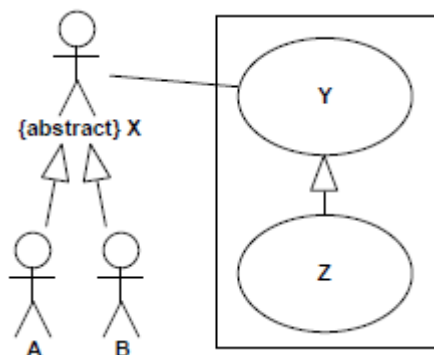
Which of the following statements about the given diagram clipping are true?



- ☐ A. A inherits all of B's associations.
- ☐ B. A can execute the same use cases as B.
- ✓ ✓ ☒ C. B can execute the same use cases as A.
- ☒ D. B inherits all of A's associations.

Câu 8 (Thời gian trả lời: 00:55)

Which of the following statements about the given diagram clipping are false?

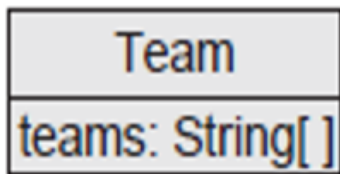


- ☐ A. Z inherits from Y, it is a specialized Y and can be executed by sub actors (descendants) of X. A and B inherit the behavior of X, they are specialized X and can execute Y.
- ☐ B. A and B can execute Y separately.
- ✗ ☒ C. A or B can execute Z.

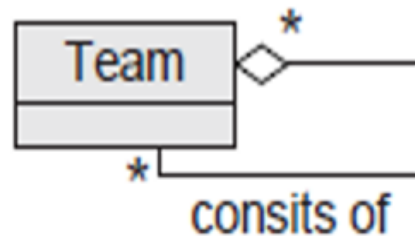
☒ D. A and B execute Y together.

Câu 9 (Thời gian trả lời: 01:05)

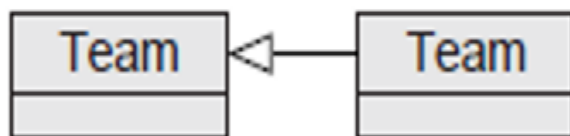
How do you model the following situation with a UML class diagram: One team may consist of several other teams.



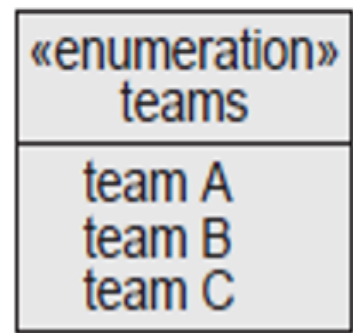
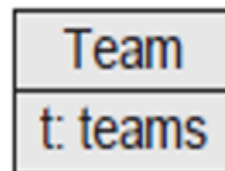
A



B



C

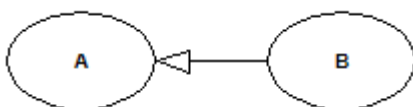


D

- ☒ A. C
☒ B. B
☐ C. A
☐ D. D

Câu 10 (Thời gian trả lời: 00:45)

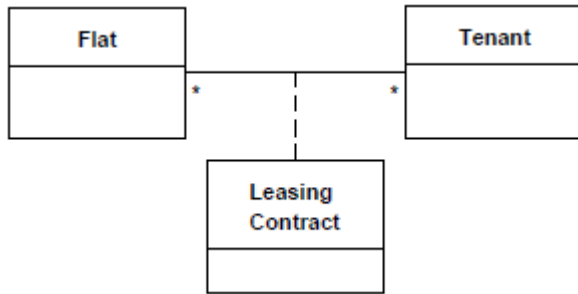
Which of the following statements about the given diagram clipping are true?



- ☐ A. B is called "base use case".
☐ B. A must not be abstract.
☐ C. A inherits all relationships from B.
☒ D. B inherits all relationships from A.

Câu 11 (Thời gian trả lời: 01:20)

You are given the following clipping of a UML class diagram. Which of the following statements are true?



- ✓ ✓ **A.** One flat can be rented by multiple tenants with different leasing contracts.
- B. One tenant rents exactly one flat.
- ✓ ✓ **C.** One tenant can rent multiple flats with different leasing contracts.
- D. One tenant can rent the same flat several times with different leasing contracts.

Câu 12 (Thời gian trả lời: 00:55)

Which of the following statements are FALSE about Use Cases?

- ☐ A. Use cases provide the basis of communication between sponsors and developers in planning phase
- ☐ B. Use cases description provides a good source to identify domain concepts
- ✓ ☒ **C.** A fully-dressed use case should include both "whats" and "hows" so that they are ready for "realization"
- ☐ D. A use case is an interaction between a user and a system

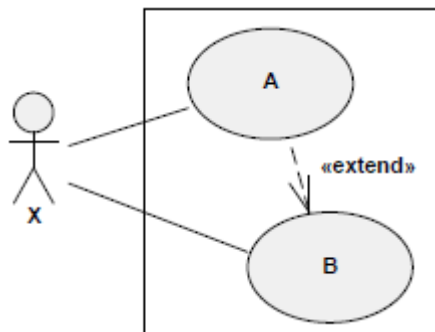
Câu 13 (Thời gian trả lời: 00:45)

Which statements are false for a use case?

- ☐ A. a use case may have many scenarios
- ☐ B. good source for identifying use cases is external events
- ✓ ☒ **C.** a use case captures some user-visible and non-visible functions
- ☐ D. a use case can be traced to a discrete goal

Câu 14 (Thời gian trả lời: 01:30)

The following <<extend>>-relationship means, that ...

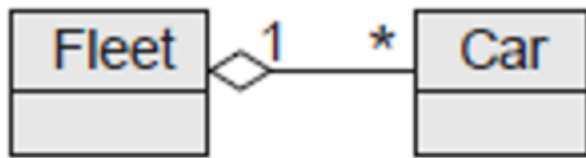


- ✓ ☒ **A.** ... A can extend B.
- ☐ B. ... A cannot be executed without B.
- ☐ C. ... B can extend A.

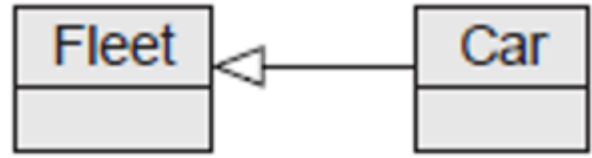
☐ D. ... B always has to invoke A.

Câu 15 (Thời gian trả lời: 00:50)

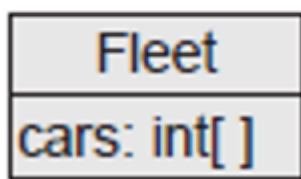
How do you model the following situation with a UML class diagram: The car fleet of a car rental contains multiple cars, one car belongs to exactly one car fleet.



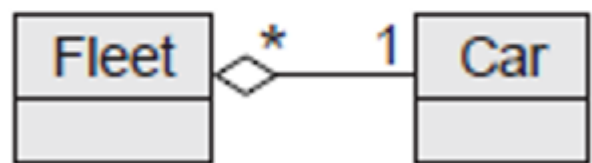
A



B



C



D

☐ A. D

☒ B. A

☐ C. B

☒ D. C

Câu 16 (Thời gian trả lời: 01:00)

What do you think you can show in a class diagram?

- a) Groups of classes
- b) The relationships between classes
- c) The dynamic behavior of classes
- d) The sequence of steps needed to complete a use case

☐ A. a, c and d

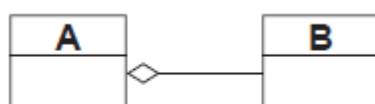
☐ B. a, b, c and d

☒ C. a and b

☒ D. a, b and c

Câu 17 (Thời gian trả lời: 01:30)

Which of the following statements about the given diagram are true?



A. If an instance of B is deleted, all contained instances of A are also deleted.

B. If an instance of A is deleted, all contained instances of B are also deleted.

✓ ✓ C.

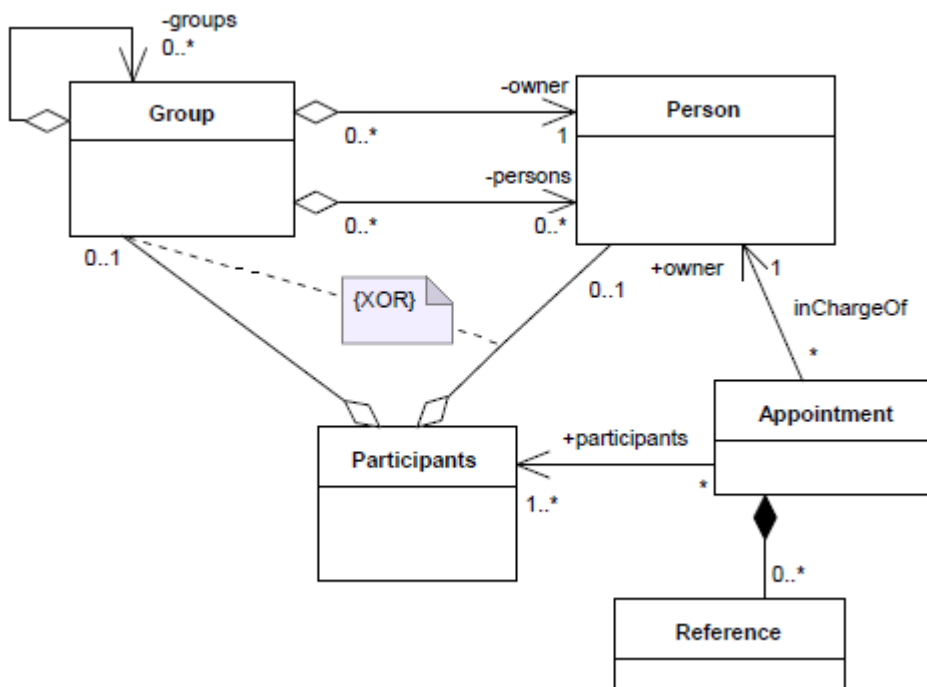
B is part of A.

✓ ✓ D. If an instance of A is deleted, the contained instances of B are not affected.

E. A is part of B.

Câu 18 (Thời gian trả lời: 00:55)

You are given the following clipping of a UML class diagram. Which of the following statements are true?



☐ A. It is possible to navigate from a person to the appointments that he is in charge of.

☒ B. A participant that has been assigned to an appointment can be group and person at the same time.

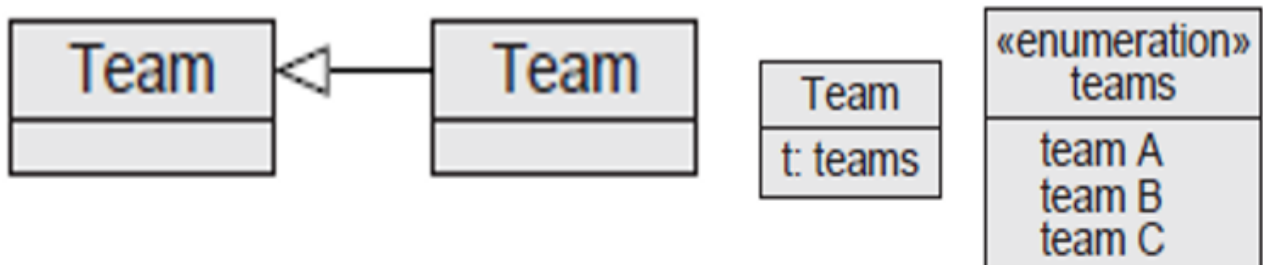
☒ C. If an appointment is deleted all references linked with it are deleted as well.

☐ D. If a group is deleted all persons that are in that group are deleted as well.

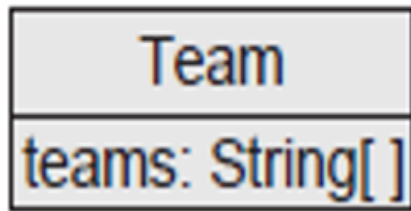
Câu 19 (Thời gian trả lời: 01:10)

How do you model the following situation with a UML class diagram:

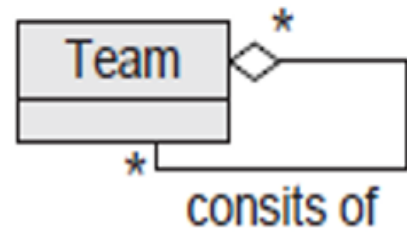
One team may consist of several other teams.



A



B



C

D

☐ A. A

☐ B. B

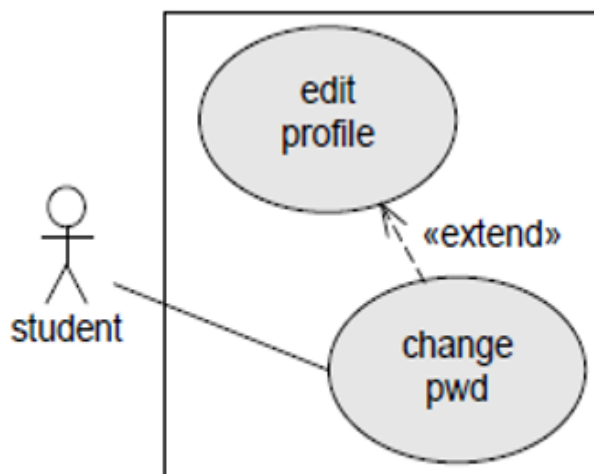
☐ C. C

☒ D. D

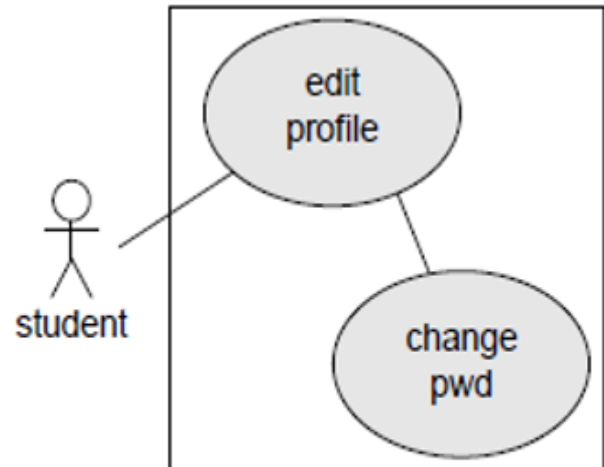
Câu 20 (Thời gian trả lời: 01:00)

How do you model the following situation with a UML2 use case diagram:

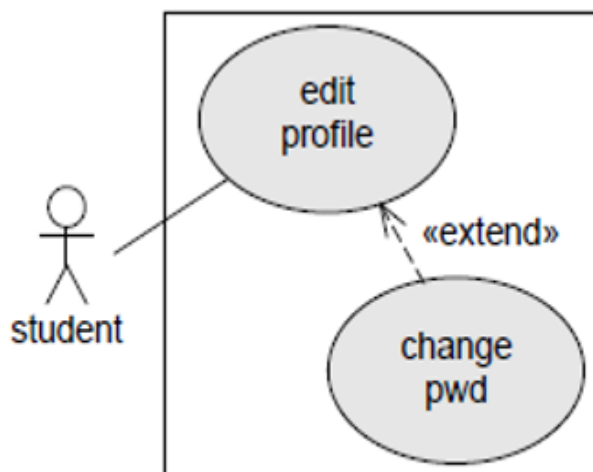
A student edits his user profile. In the course of that he can also change his password if he likes.



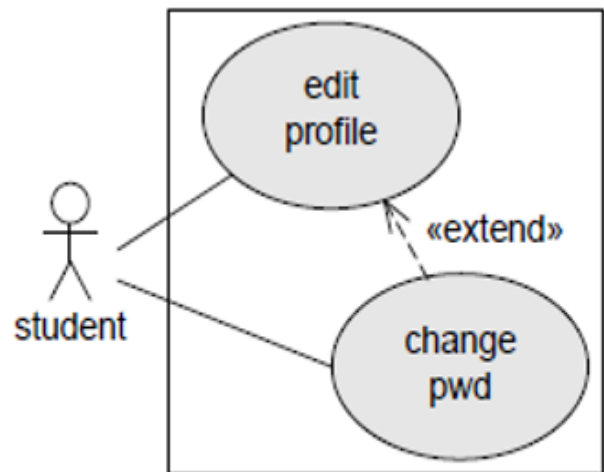
A



B



C

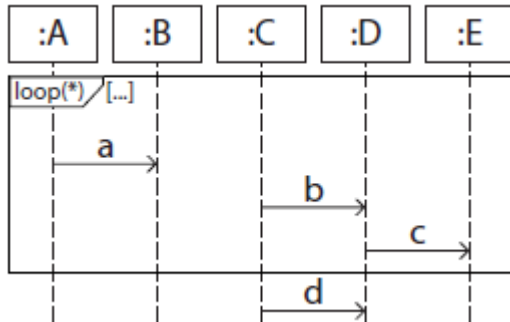


D

- ☒ A. D
☒ B. C
☐ C. A
☐ D. B

Câu 21 (Thời gian trả lời: 01:05)

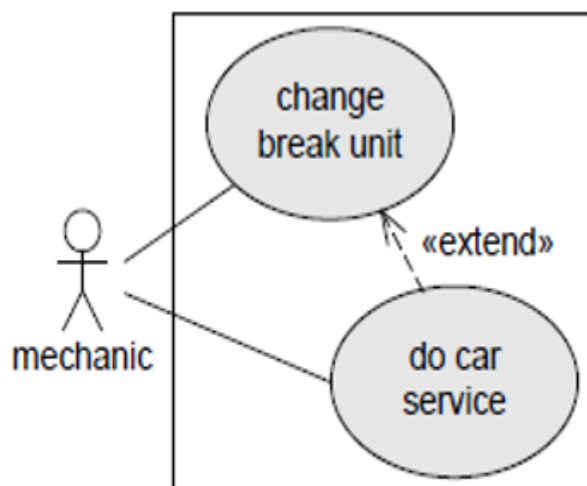
You are given the following sequence diagram. Which traces are impossible?



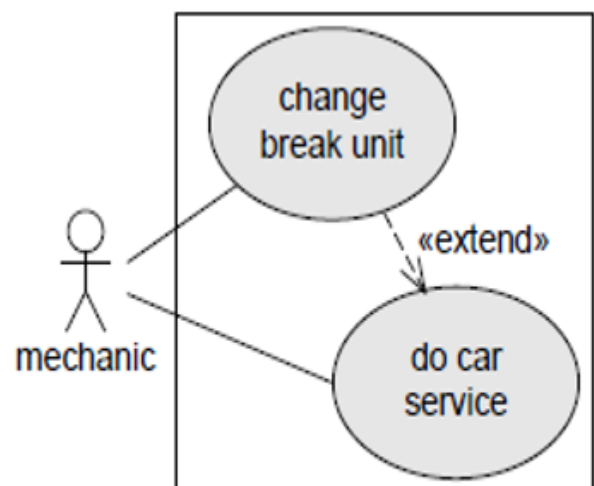
- ☐ A. $b \rightarrow c \rightarrow a \rightarrow a \rightarrow b \rightarrow c \rightarrow d$
☐ B. $a \rightarrow b \rightarrow c \rightarrow d$
☒ C. $d \rightarrow a \rightarrow b \rightarrow c$
☒ D. $b \rightarrow a \rightarrow c \rightarrow d$

Câu 22 (Thời gian trả lời: 01:05)

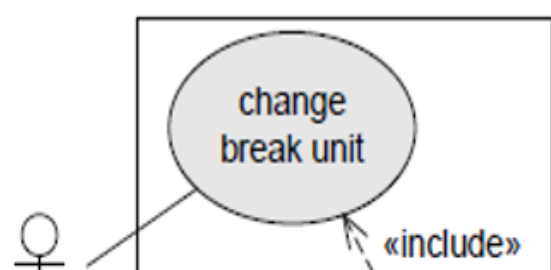
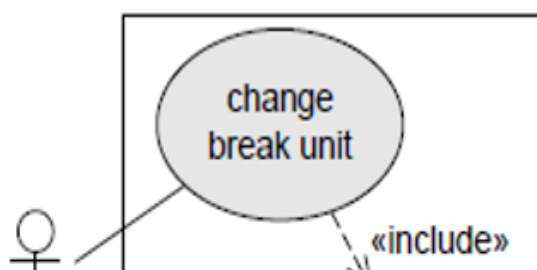
How do you model the following situation with a UML use case diagram: A mechanic does a car service. During that service, it might be necessary to change the break unit.

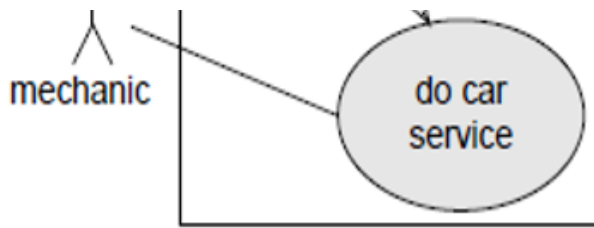


A

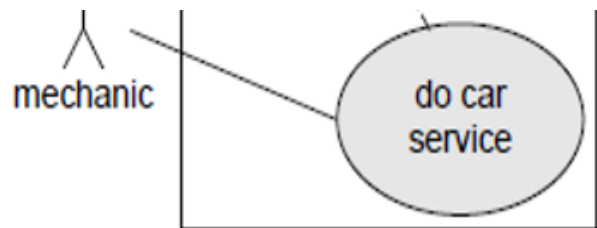


B





C



D

- ☐ A. D
- ☐ B. C
- ☒ C. B
- ☒ D. A

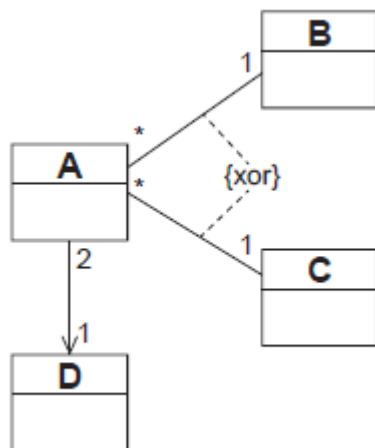
Câu 23 (Thời gian trả lời: 00:45)

Within sequence diagrams ...

- ☒ A. you can refer to other sequence diagrams
- ☐ B. the behavior of one object is modeled.
- ☐ C. ... time is no separate dimension.
- ☒ D. the chronological process of one object is modeled.

Câu 24 (Thời gian trả lời: 00:55)

You are given the following clipping of a UML class diagram. Which of the following statements are false?

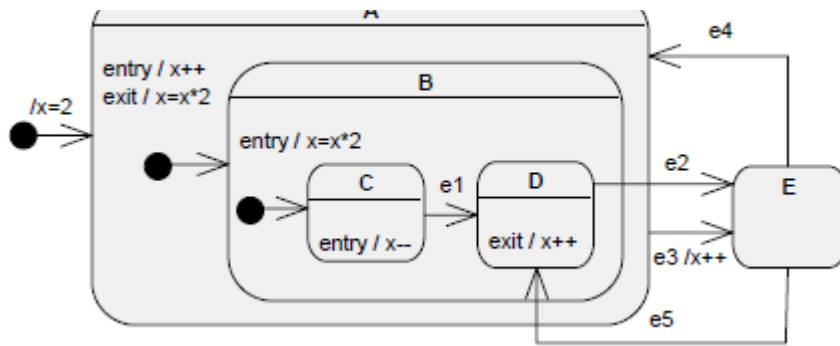


- ☐ A. D
- ☒ B. C
- ☒ C. B
- ☐ D. A

Câu 25 (Thời gian trả lời: 01:30)

You are given the following state machine diagram. What is the value of x after the occurrence of the event chain

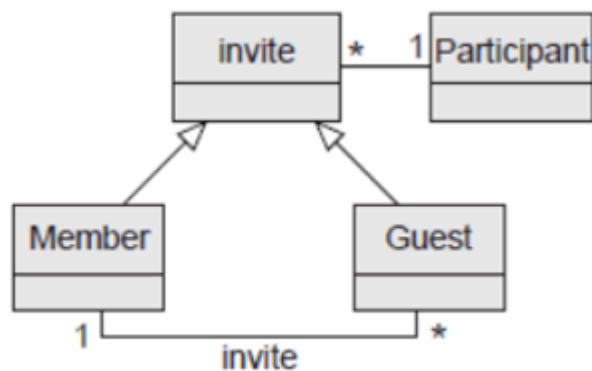
e1 e2 e4 e2 e1?



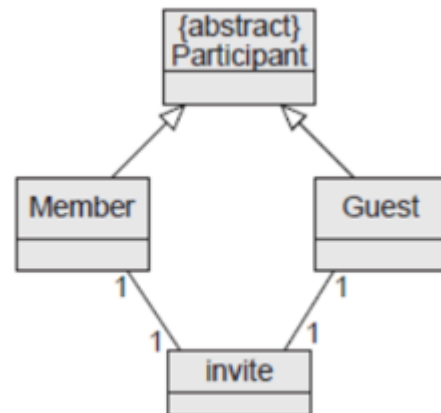
- ☐ A. 26
☐ B. 12
☒ C. 25
☐ D. 13

Câu 26 (Thời gian trả lời: 01:15)

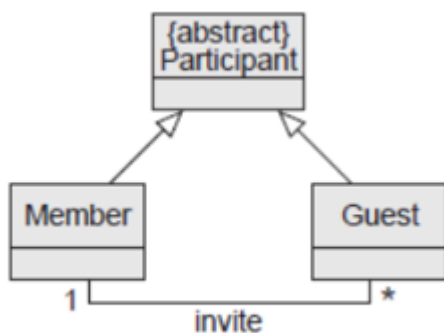
How do you model the following situation with a UML class diagram: There are exactly two kinds of participants, namely members and guests. Guests are invited by members.



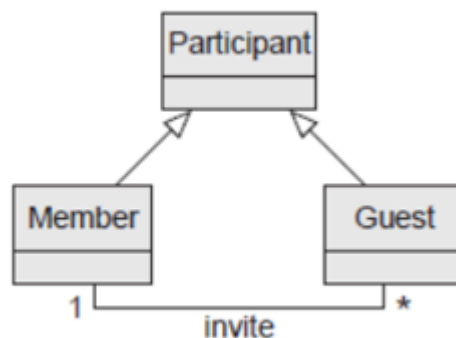
A



B



C

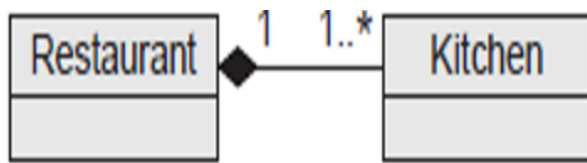


D

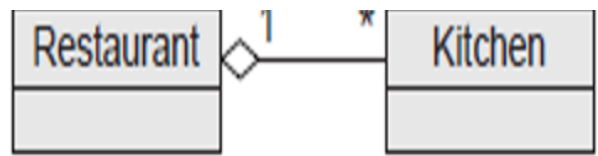
- ☒ A. C
☐ B. A
☐ C. B
☐ D. D

Câu 27 (Thời gian trả lời: 01:00)

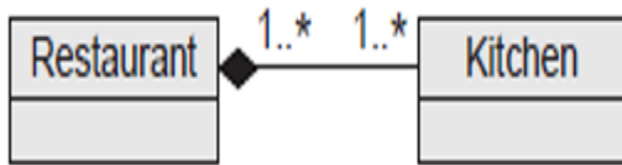
How do you model the following situation with a UML class diagram: Every restaurant has at least one kitchen, one kitchen is part of exactly one restaurant.



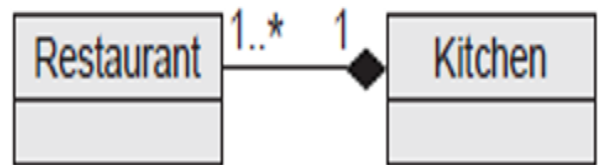
A



B



C



D

- ☒ A. C
☐ B. D
☐ C. B
☐ D. A

Câu 28 (Thời gian trả lời: 00:45)

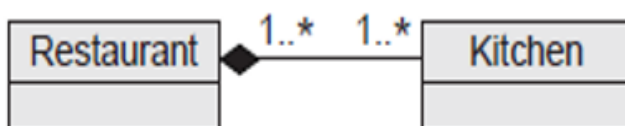
Associations between classes are implemented by ... ?

- ☒ A. function calls
☐ B. types
☐ C. functions
☐ D. variables

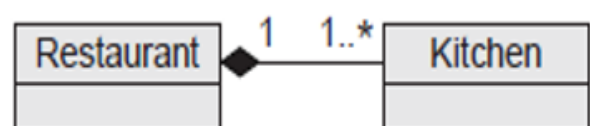
Câu 29 (Thời gian trả lời: 01:10)

How do you model the following situation with a UML class diagram:

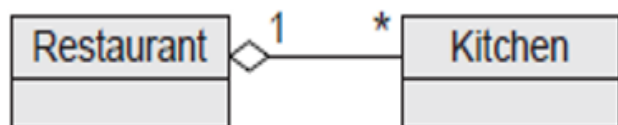
Every restaurant has at least one kitchen, one kitchen is part of exactly one restaurant.



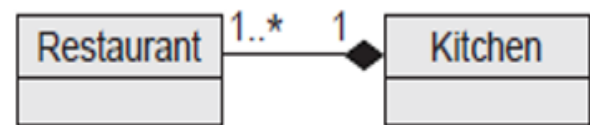
A



B



C



D

- ☐ A. B
☒ B. A
☐ C. C
☐ D. D

Câu 30 (Thời gian trả lời: 00:50)

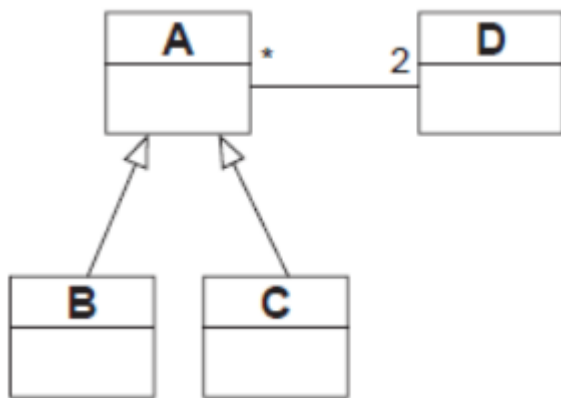
Which of the following combine to form an OOAD method?

- a) Notation/Formal specification
- b) Diagram
- c) Process
- d) View

- ☒ A. a, b and d
- ☐ B. a and b
- ☐ C. a and c
- ☒ D. a, b and c

Câu 31 (Thời gian trả lời: 01:05)

You are given the following clipping of a UML 2 class diagram. Which of the following statements are FALSE?



- ☒ A. One object of D may be associated with multiple objects of C.
- ☐ B. There are no direct instances of A.
- ☒ C. Two objects of D are associated with * objects of A.
- ☐ D. One object of B must be associated with exactly two objects of D.

Câu 32 (Thời gian trả lời: 00:50)

Which of the following statements are true?

- ☐ A. None of the above.
- ☒ B. Maintaining different models for different perspectives will run into a risk of inconsistency among models.
- ☐ C. As models are refined, it is acceptable to use the same set of class diagrams for different perspectives: conceptual, specification and implementation.
- ☐ D. All of the above.

Câu 33 (Thời gian trả lời: 00:50)

Which statements are false for a use case?

- ☐ A. a use case may have many scenarios

- ☐ A. a use case may have many scenarios
- ☐ B. good source for identifying use cases is external events
- ☐ C. a use case must be initiated by an actor
- ✓ ☒ D. a use case captures some user-visible and non-visible functions

Câu 34 (Thời gian trả lời: 00:55)

A sequence diagram is a two-dimensional chart. The two dimensions are ...

- ☐ A. vertical: roles / horizontal: time axis
- ✓ ☒ B. vertical: time axis / horizontal: roles
- ☐ C. other answer

Câu 35 (Thời gian trả lời: 00:55)

Which of the following statements are FALSE about Use Cases?

- ☐ A. Use cases provide the basis of communication between sponsors and developers in planning phase
- ✓ ☒ B. A fully-dressed use case should include both "whats" and "hows" so that they are ready for "realization"
- ☐ C. Use case diagrams are the primary tool to document requirements
- ☐ D. Use cases description provides a good source to identify domain concepts

Câu 36 (Thời gian trả lời: 01:15)

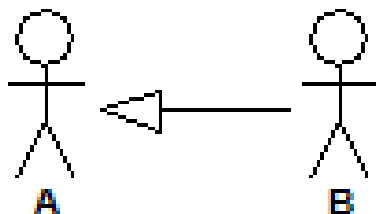
In order to model the relationship “a course is composed of 5 to 20 students and one or more instructors”, you could use:

- a) To model systems using object-oriented concepts
- b) To provide a process for software development
- c) To support small-scale and large-scale analysis and design
- d) To provide an insight into implementation mechanisms

- ✗ ☒ A. Association
- ☐ B. Composition
- ☐ C. Realization
- ☐ D. Aggregation

Câu 37 (Thời gian trả lời: 01:30)

Which of the following statements about the given diagram clipping are true?



- ✓ ✓ A. B can execute the same use cases as A.
- B. A inherits all of B's associations.

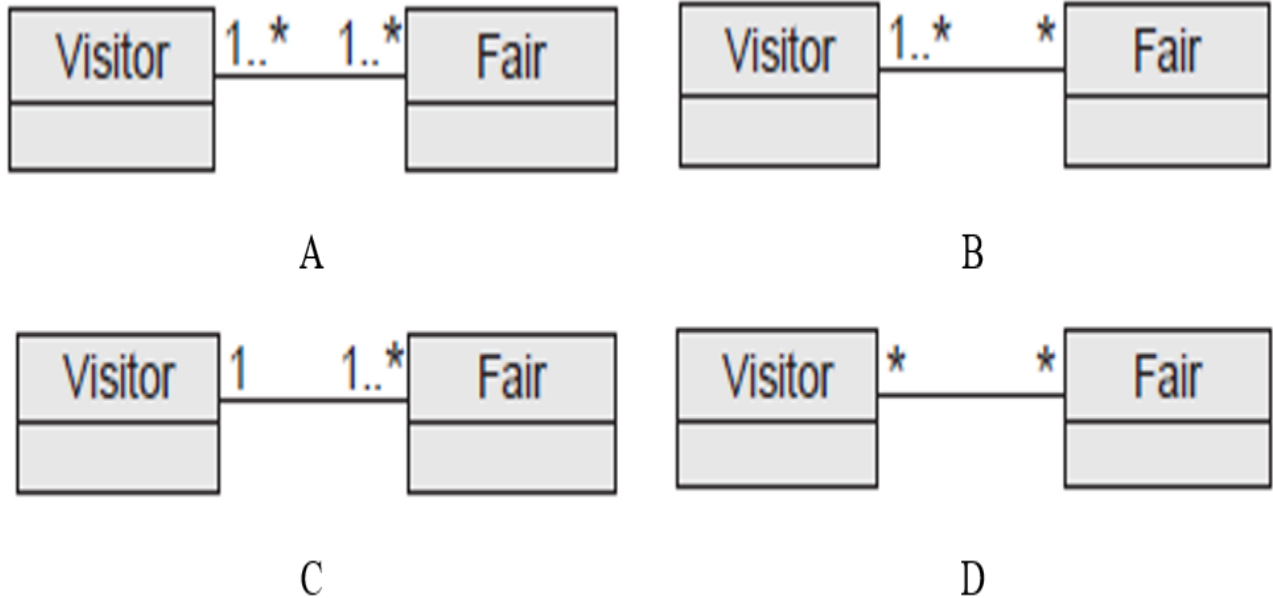
C. A can execute the same use cases as B.

✓ ✓ D. B inherits all of A's associations.

Câu 38 (Thời gian trả lời: 01:00)

How do you model the following situation with a UML class diagram:

A fair is visited by at least one visitor. One visitor visits at least one fair.



- ☒ A. A
- ☐ B. B
- ☐ C. C
- ✗ ☒ D. D

Câu 39 (Thời gian trả lời: 00:50)

When to use Interaction Diagrams?

- ☐ A. They are good to look at behavior of a single object across many use cases
- ☐ B. When you want to look at the behavior of several objects within a single use case or several use cases
- ☐ C. All of the above
- ✓ ☒ D. They are good at showing collaborations among objects; they are not so good at conditions and looping

Câu 40 (Thời gian trả lời: 01:05)

Which statements are false for a use case?

- ☐ A. a use case may have many scenarios
- ✓ ☒ B. UML use case diagrams are designed to replace textual description since a picture is better than a thousand words
- ☐ C. a use case must be initiated by an actor
- ☐ D. a use case can be traced to a discrete goal

